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Optimal site selection for utility-scale solar PV projects using a RETscreen-AHP-TOPSIS framework: application to the southern Cameroon

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Abstract

Several countries have renewable energy portfolios aimed at reducing energy sector emissions while diversifying their generation mixes. A cornerstone in these global portfolios is solar photovoltaics (PV) which offer a mature and rapidly deployable solution. Selecting optimal sites for utility-scale solar PV projects remains a complex challenge, requiring the integration of technical, economic, environmental and social criteria. This study proposes a hybrid decision-making framework that embeds financial viability metrics such as net present value (NPV) and greenhouse gas (GHG) emissions savings into multi-criteria analysis. The framework integrates RETScreen Expert for technoeconomic evaluation, the analytical hierarchy process (AHP) for criteria weighting, and the Technique for Order Preference by Similarity to Ideal Solutions (TOPSIS) for ranking alternatives. Applied to fourteen locations in the Southern Interconnected Grid in Cameroon, the model considers a hypothetical 10MW grid-connected solar PV plant. Results identify Baffousam as the optimal location due to its higher solar potential, favorable land use, and grid proximity. Sensitivity analysis reveals the influence of subjectivity weighting in AHP, with land use criterion contributing significantly to site selection, underscoring the need for stakeholder engagement. By bridging traditional energy planning tools (RETScreen) with MCDM techniques, this study addresses a critical gap in existing frameworks which often neglect the time value of money and financial viability. The modular design allows adaptation to regional priorities, offering policy makers and investors a replicable blueprint for emerging economies. This approach advances sustainable energy transitions while balancing technical, economic, and environmental priorities in site selection.

Keywords Renewable energy, Cameroon, RETScreen, Analytic hierarchy process, Technique for order preference by similarity to ideal solutions

Introduction

It is axiomatic that energy demand, particularly modern energy, is directly correlated to economic growth. The world's energy supply has evolved from traditional biomass in the pre-industrial era, to coal in the industrial revolution, and to the current mix of coal, petroleum, nuclear and renewable energy sources. Of the various energy consumption types, electricity is arguably the most important. Electricity is versatile, indispensable to our modern life, acts as the backbone of technological advancement, and is a vital commodity in the current

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global economy. The three main categories of sources for electricity generation are renewable energy sources (RES), nuclear energy, and fossil fuels (United States Energy Information and Administration 2022). The environmental impacts associated with nuclear energy and fossil fuels, coupled with the unsustainability of the latter have seen the focus shift to RES (IEA, 2021). RES includes geothermal, solar photovoltaics (PV), wind, energy storage, and controllable loads such as like electric vehicles (IRENA & REmap, 2030) (IRENA, 2020). The recent upsurge in the adoption of RES is also attributed to the drastic decline in costs, especially proven technologies. Solar PV, onshore and offshore wind which experienced costs reduction of 82%, 40% and 29% respectively in the last decades (IRENA, 2019).

Despite the rising costs of key materials needed for their manufacture, electricity generation from RES (mainly from solar and wind) is expected to accelerate in the coming years (Carrara et al., 2020). For example, global installed capacity for renewables is forecasted to grow by over 60% from 2020 levels, to about 4 800 GW by 2026 (IEA, 2021). However, this forecasted growth in RES is mainly expected in the Global North which have already attained satisfactory levels of electricity access, in contrast to countries in Africa and the Global South (Sharma et al., 2019). Access to electricity in Africa was 56% in 2019. This represented an estimated 579 million people without access to electricity. Electricity generation from RES in the continent contributes only about 3% to global electricity generation from RES, despite the continent having the highest solar resource potential on the planet (IEA, 2019) (IRENA, 2022). Although the continent has made considerable progress in advancing access to electricity, expansion rates have been outpaced by rate of population growth, particularly in sub-Saharan Africa (IRENA, 2022). Access to electricity in sub-Saharan Africa was 48%, with an estimated 578 million people without electricity access. The lack of reliable, affordable and modern energy hinders industrial development, agricultural productivity, delivery of crucial public services (such as education, health care), and general sustainable development of the region (IRENA, 2022) (Vezzoli et al., 2018).

The peculiarities of sub-Saharan countries, in terms of lower population densities, political and structural deficiencies in their economies, and dilapidated nature of their power sector necessitate innovative approaches to increasing electricity access (Urban et al., 2007). Amid the increasing energy demand, RES is principally positioned to increase national generation capacities given their abundant and naturally dispersed nature in the region. RES would also catalyze the socio-economic development and industrialization of the sub-Saharan

Africa, in addition to hedging against economy impacts of fossil fuel price volatility (Failed, 2013). Many nations including Cameroon have RES portfolios and targets, aimed at enabling a more secure, diversified, sustainable and low-carbon emission energy sector. Solar PV is the mainstay in these RES portfolios especially in sub-Saharan Africa due to their lowest lead time, minimum technology requirements, minimum costs and widest public acceptance (IRENA, 2019).

Utility-scale PV plants, generally defined as solar PV plants with at least 5MW of installed capacity, arguably hold the most potential for harnessing solar energy (Al Garni & Awasthi, 2016). Their deployment requires vast amounts of land and areas with high average yearly solar irradiation to mitigate the variability of solar resource. Determining the location of a utility-scale PV plant requires the examination of certain aspects such as viability of the site and proximity to existing infrastructures. The location goal is centered on maximizing energy generation while minimizing the total project costs over the lifetime of the project. Carrying out a comprehensive solar PV site assessment is a vital phase towards ensuring effective and profitable utility-scale PV project. Traditional evaluation methods use energy planning tools which are often single criterion focused, such as maximizing energy generation. Considering the various criteria that influence site selection for utility-scale PV projects, multi criteria decision-making (MCDM) approaches/methods have been increasingly adopted to overcome the limitations of these traditional tools.

Cameroon aims for renewable energy to constitute 25% of its generation mix by 2035, with solar PV exceeding 6%—a share expected to rise given the country's growing preference for solar PV over other renewable sources (Ayuketah et al., 2024) (Iweh et al., 2023a) (Ayuketah et al., 2023a). The decision-making process for renewable energy projects, including utility-scale PV site selection is often divided into three phases (Al Garni & Awasthi, 2016) (Noorollahi et al., 2007): identification of evaluation criteria and exclusion aspects for the potential sites; model-based evaluation and ranking of potential sites; and sensitivity analysis to determine the robustness of the process. Given the numerous criteria to be considered, many studies have applied both traditional and MCDM approaches, including recent studies incorporating Geographical Information System (GIS) and fuzzy logic as reviewed in Kumar et al. (2017) (Shao et al., 2020) (Campos-Guzmán et al., 2019) (Al Garni & Awasthi, 2017) (Yimen & Dagbasi, 2019) (Akçay & Atak, 2018) (Rekik & Alimi, 2024a) (Failed, 2023).

However, the selection of optimal sites for utility-scale solar PV projects in sub-Saharan Africa, particularly Cameroon, faces critical gaps in existing methodologies.

While traditional energy planning tools such as HOMER excel in technoeconomic analysis, they often neglect multi-criteria-decision-making frameworks that account for conflicting technical, economic, environmental and social factors (Kumar et al., 2017) (Shao et al., 2020). Conversely, standalone MCDM methods lack integration with financial and environmental viability metrics (such as net present value (NPV) and greenhouse gas (GHG) emissions savings respectfully), limiting their realism in project feasibility assessments. Additionally, prior studies in Cameroon and similar regions overlook the socio-economic impacts of land use, a critical oversight in contexts where land conflicts and competing uses are prevalent.

This study addresses these gaps by proposing a novel hybrid RETScreen-AHP-TOPSIS framework, the first of its kind applied to Cameroon. Unlike prior works, the framework integrates RETScreen's financial metrics (energy yield, NPV, GHG savings) with MCDM methods, bridging the divide between project viability analysis and multi-criteria-decision making. It uniquely quantifies land use criteria, offering insights into decision-making robustness in resource constrained settings. The contributions of this work are threefold: a replicable hybrid framework that embeds the time value of money into MCDM, enhancing decision realism for utility-scale solar PV projects. Furthermore, it provides context specific insights by ranking optimal sites (the best being Baffousam) in the southern grid of Cameroon, addressing the region's unique challenges (land use conflicts, grid accessibility and economic constraints). Additionally, its policy relevant as it gives direct support to Cameroon's 2035 renewable energy target and Nationally Determined Contributions (NDCs) by informing solar PV deployment strategies and fostering stakeholder consensus through transparent sensitivity analysis (Ayuketah et al., 2024) (Ayuketah et al., 2023a).

The remainder of this paper is structured as follows. Section "Literature review" presents the selected evaluation criteria, and reviews traditional energy planning tools and modern MCDM approaches for renewable energy site selection, highlighting their applications. Section "Methodology" details the proposed RETScreen-AHP-TOPSIS methodology, including an overview of RETScreen's techno-economic analysis, AHP's criteria weighting, and TOPSIS's ranking mechanisms. Section "Case study" presents a case study for southern Cameroon, numbering the criteria, alternative sites and data sources. Section "Results and Discussion" presents and discusses the results, including optimal site rankings, sensitivity analysis, and actionable insights for policy makers. It also identifies limitations and future research

directions, such as dynamic modelling. Finally, Section VI concludes by giving a summary and policy directions of the study,

Literature review

Decision criteria and selection factors

Depending on the objective, several criteria influence the location of utility-scale solar PV projects. A comprehensive review of these criteria is provided in Kumar et al. (2017) (Shao et al., 2020) (Kaya & Kahraman, 2011). According to Shao et al. (2020), these criteria can be categorized into two: exclusion and evaluation criteria. Exclusion (or restriction) criteria are used to identify and eliminate sites unsuitable for locating solar PV plants, such as protected areas and urban settlements. Conversely, evaluation criteria are utilized for appraising and ranking potential site locations that do not satisfy restriction criteria. Evaluation criteria include costs, distance to roads and public security. Evaluation criteria for solar PV plants, reviewed in Shao et al. (2020) (Kaya & Kahraman, 2011) (Deveci et al., 2021) (Wang et al., 2018a) are often grouped into technical, economic, environmental and social factors. Selection of criteria.

The evaluation criteria considered in this study were selected based on reviewed literature and applicability. They include:

Technical criteria

- *Energy generated (C1)*: This is the amount of energy generated by the utility-scale PV project over a specified period, often 1 year.
- *Capacity factor (C2)*: The ratio of the energy generated by the solar plant over a given period compared to maximum theoretical energy generation.
- *Proximity to residential areas (C3)*: Utility-scale solar PV plants are usually located not too far, nor near residential areas. The former is to ensure energy is generated as close as possible to load to minimize transportation losses or costs. The latter aims to mitigate the health-related impacts of local population as well as minimize security issues relating to unauthorized visits to project sites. Optimal locations from residential areas are often sought. A buffer of 100 m was considered.

Economic criteria

- *Proximity to roads (C4)*: This mainly influences the logistics aspects during project site development. Constructing utility-scale solar PV plants involves the mass transportation of solar panels, and other

associated equipment to project sites. Thus, the identified project area should be cost-competitively accessible both during the construction and operation phases. The latter is necessary for the maintenance team during planned or unplanned activities.

- *Proximity to transmission lines (C5)*: The amount of power loss is directly proportional to the characteristics of the transmission line. It's important to site solar PV projects close to transmission or distribution lines as possible to minimize energy losses and maximize profits. Furthermore, the distance to evacuation lines significantly influences the economics of the project, given transmission infrastructure will have to be built thereby increasing the capital costs. Short distances are often preferred
- *Net present value (NPV) (C6)*: It expresses the present costs of future cash flows over the lifecycle of the utility-scale PV project, discounted to the present cost
- *Simple payback period (SPP) (C7)*: This shows how long it takes the utility-scale PV project to recoup on the initial investments. It permits the comparison of alternative sites, with the shortest time (usually years) often favored.
- *Land use (C8)*: Given utility-scale solar PV plants require vast areas of surface land, the present and planned future use of potential location are vital to project viability. Land availability and land use such as forest, cultivated land and for pasture often contribute directly to project costs. This has an impact on the economic aspect, since the project land must be legally safe against extra cost elements or outside interference during the project.

Environmental criteria

- *Proximity to protected areas (C9)*: Many countries and communities have reserves, restricted or protected areas earmarked for various reasons. These areas include historical sites, planned future community projects, parks, wildlife reserves, etc. It's important to identify or delineate these areas so appropriate considerations are made during appraising project areas.
- *Proximity to water ways (C10)*: Siting utility scale PV projects close to waterbodies have double effects. Variation in water vapor could lead to a reduction in the incident radiation on the panel, subsequently reducing the output power (Guechi et al., 2011). However recent literature indicates an inverse effect. Water droplets on the PV panel cool the panel,

decreasing the panel temperature leading to a higher potential difference and an overall increase in panel power output (Mustafa et al., 2020).

- *GHG emissions savings (C11)*: GHG emissions of electricity generation plants vary depending on the technology used. Considering that solar PV is largely GHG emission free, this criterion assesses the avoided emissions if the same energy were to be generated on the current generation mix in the system.

Decision making methods

Selecting an optimal location for utility-scale solar PV projects among several alternatives presents a complex decision-making challenge due to the multitude of technical, economic, environmental and social criteria that must be considered. Traditional approaches to addressing this problem often rely on energy planning tools such as Hybrid Optimization Model for Electric Renewables (HOMER), RETScreen Expert, PVSyst and System Advisor Model (SAM) as reviewed in Tozzi and HoJo (2017) (Ringkjøb et al., 2018). For instance, HOMER Pro was employed to conduct a techno-economic analysis of a hybrid solar-wind system in Puerto Bolivar, Colombia (Budes et al., 2020), and to evaluate an off-grid pumped-hydro energy storage for electrifying Djoundé village in northern Cameroon (Yimen et al., 2018). Similarly, RETScreen was utilized to assess the techno-economic and environmental sustainability of a grid-connected solar PV plant in Northeastern Nigeria (Owolabi et al., 2019), and grid connected in Yaounde- Cameroon (Iweh et al., 2023b), while another study analyzed the profitability and reliability of integrated PV systems using the same tool (Failed, 2018). PVSyst was applied to estimate energy generation of a large-scale, grid-connected solar PV plant in Bikaner, Rajasthan (Sharma et al., 2022), and the System Advisor Model (SAM) was used to evaluate the performance of concentrated solar power (CSP) plant located in Crowley, Louisiana (Ezeanya et al., 2018). However, traditional approaches have predominantly focused on single-criterion analysis, such as cost minimization or energy output maximization (Kumar et al., 2017) (Shao et al., 2020), which often fail to address the multifaceted nature of modern decision-making problems.

The increasing complexity of decision-making in renewable energy projects, driven by the multiplicity of conflicting criteria has led to the adoption of MCDM methods. These methods allow for the structured consideration of diverse preferences have become the predominant approach in recent years (Kumar et al., 2017). Unlike single-criterion methods, MCDM approaches offer the distinctive advantage of handling multiple criteria in an integrated manner, leading to a more robust

and informed decision making (Shao et al., 2020). The effectiveness of MCDM methods has spurred the development of various approaches, including the Analytic hierarchy process (AHP), Weighted Product Method, Weighted Product Method and Multi-Attribute Utility Theory (MAUT), POMETHEE (Preference Ranking Organization Method for Enrichment Evaluation), ELECTRE family (Elimination Et Choix Traduisant la Réalité), TOPSIS (Technique for Order Preference by Similarity to Ideal Solutions), VIKOR (Multicriteria Optimization and Compromise Solution), DEA (Multi-Criteria Data Envelopment Analysis). Recent studies have advanced renewable site selection using MCDM methods, combining them with GIS and Fuzzy logic approaches. Comprehensive review of MCDM methods and their applications are provided in Kumar et al. (2017) (Shao et al., 2020) (Campos-Guzmán et al., 2019) (Kaya et al., 2019) (Siksnylyte et al., 2018). These methods can be broadly categorized into value measurement models, goal, aspiration and reference level models, and outranking models. These models could also be combined with various derivatives of each method (Kumar et al., 2017).

MCDM approaches have been widely applied in renewable energy site selection, demonstrating their versatility and effectiveness. For example, fuzzy-AHP for weighting combined with fuzzy-TOPSIS was utilized to select the best alternative energy technologies in Turkey (Kaya & Kahraman, 2011) and to identify optimal wind plant locations in Vietnam (Wang et al., 2018a). In hybrid applications in Tunisia, a SWOT-SWARA-TOPSIS-G was utilized to rank solar-based green hydrogen sites (Rekik & Alimi, 2024b), evaluate and optimize green hydrogen strategies (Rekik, 2024), GIS-FAHP-FTOPSIS for optimal wind site selection (Faied, 2023), and an integrated SWARA-DEMATEL-GIS to assess solar-wind hybrid potential in the Kasserine region (Rekik & Alimi, 2023). Applications for solar PV site selection include utilizing a fuzzy DEA-AHP-TOPSIS methodology to select the best solar plant location in Vietnam (Wang et al., 2018b). GIS integrated with AHP have been applied to identify optimal solar PV locations in various contexts, such as in Saudi Arabia (Al Garni & Awasthi, 2017), the Malatya province in Turkey (Colak et al., 2020), and in Tunisia where it was integrated with FAHP (Rekik & Alimi, 2024a). Similarly, the AHP-fuzzy VIKOR approach was used to evaluate solar PV projects sites in Pakistan (Solangi et al., 2019), while the AHP-Modified Brown-Gibson Model was applied to determine suitable sites for solar PV projects in northern Cameroon (Yimen & Dagbasi, 2019). In Spain, AHP was used for criteria weighting, and TOPSIS was applied to evaluate alternative locations for solar PV plants in Cartagena (García-Cascales & Sánchez-Lozano, 2013). Additionally,

AHP-TOPSIS was employed to assess the feasibility of solar farm deployment in India (Sindhu et al., 2017).

The selection of AHP and TOPSIS as MCDM tools for this study is due to their complementary strengths in addressing complex decision-making problems in solar PV site selection, AHP is highly effective for structuring decision-making criteria, determining their relative weights through pairwise comparisons and ensuring consistency in judgments (Al Garni & Awasthi, 2017). TOPSIS on the other hand excels in ranking alternatives based on their proximity to an ideal solution, providing clear and intuitive approach to decision-making (Akçay & Atak, 2018) (Nazari et al., 2018). Unlike other classical approaches such as PROMETHEE and ELECTRE which involve complex outranking procedures, AHP and TOPSIS offer simplicity and transparency, making them more accessible for practical applications. Compared to fuzzy approaches like FAHP and FTOPSIS, AHP and TOPSIS are less computationally intensive and do not require the additional complexity of fuzzy logic which is primarily used to handle uncertainty and imprecision. Furthermore, Grey techniques such as Grey Relational Analysis (GRA) are useful for incomplete data but lack the structure weighting framework of AHP and the clear ranking mechanisms of TOPSIS. Thus, AHP and TOPSIS are preferred for their balance of simplicity, transparency, effectiveness in handling the multi-criteria decision problems, particularly for utility scale- solar PV site selection (Kumar et al., 2017) (Shao et al., 2020) (Kaya et al., 2019) (Siksnylyte et al., 2018) (Huang & Yoon, 1981) (Mateo, 2012).

Methodology

A four-stage procedure is developed to identify and rank optimal sites for utility-scale solar PV projects in Cameroon. The framework, depicted in Fig. 1 integrates GIS, RETScreen, and MCDM methods (AHP-TOPSIS) to address the technical, economic and environmental criteria systematically. The stages are summarized as follows:

- Step 1: Determination of evaluation and restriction criteria

From the comprehensive literature review, restriction criteria (e.g., protected areas, water bodies urban zones) and evaluation criteria (e.g., irradiance, grid proximity, and land use) related to utility-scale solar PV plants are identified. The criteria to be considered are also selected. Additionally, the alternative locations chosen within the geographic area of the Southern Grid are either regional capital towns, or towns with a population exceeding 100,000.

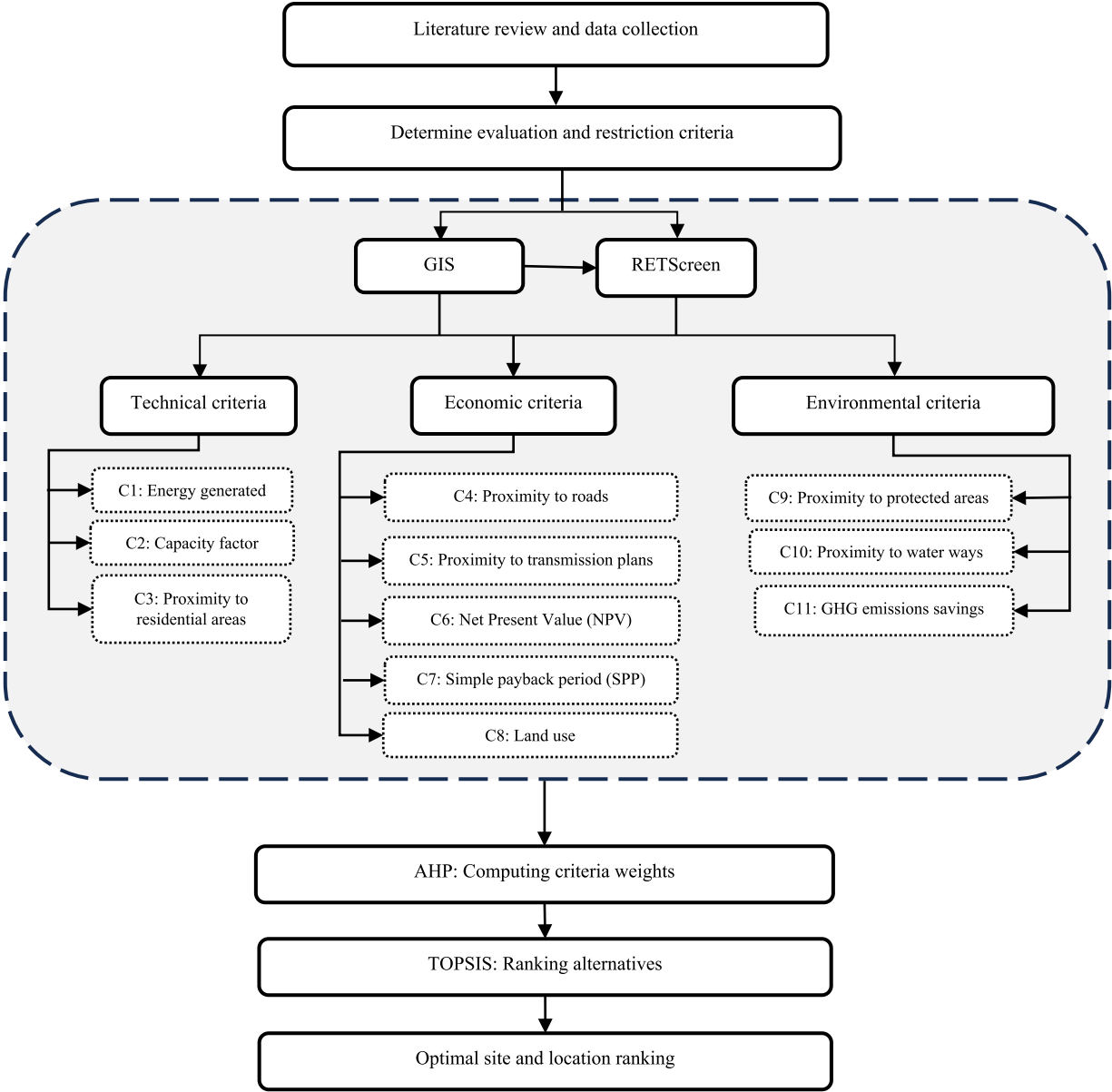


Fig. 1 Proposed framework for optimal siting of utility-scale solar PV locations in Cameroon

- Step 2: GIS-based site screening and RETScreen analysis

For each town, GIS was utilized to identify candidate sites while ensuring compliance with restriction criteria. At each site, spatial attributes (such as Proximity metrics) and geographic coordinates (latitude/longitude) for climate data retrieval are recorded. Alongside financial assumptions, the latter is fed into RETScreen which computes techno-economic metrics (NPV, GHG

savings, capacity factor). This step results in the generation of the complete set of technical, economic, and environmental criteria.

- Step 3: AHP-based criteria weighting

Here, AHP assigns weights to criteria through pairwise comparisons, and then validated for consistency. This step quantifies stakeholder priorities and mitigates subjectivity biases.

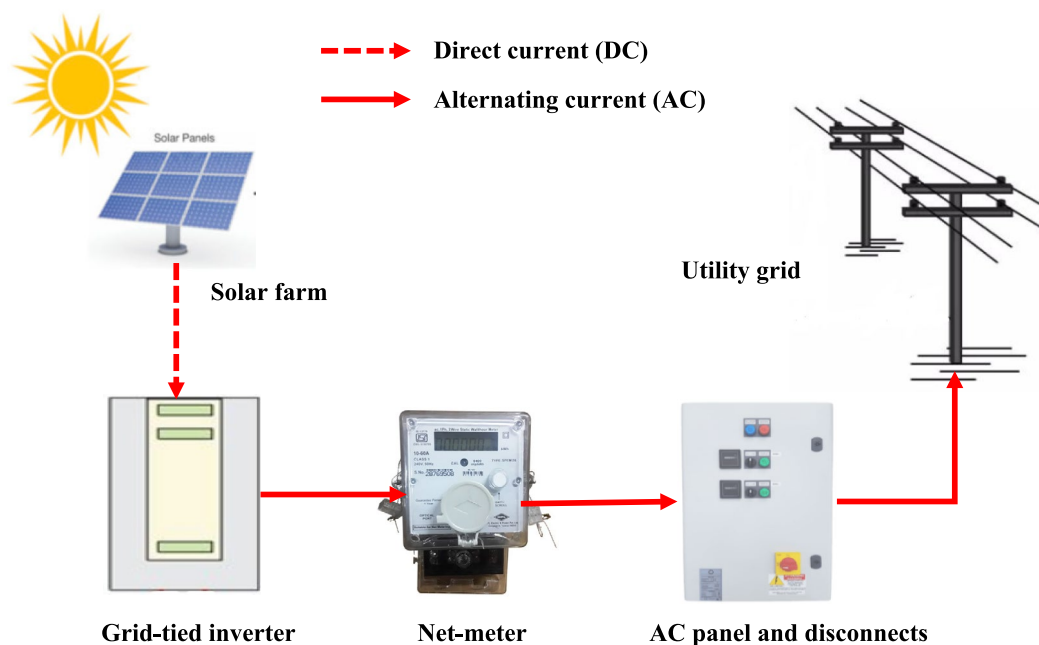


Fig. 2 Configuration of the proposed utility solar PV system

- Step 4: TOPSIS ranking of alternatives

Alternative sites are then ranked by normalizing criteria values, computing proximity to ideal/anti-ideal solutions, and generating preference scores, with higher scores indicating greater suitability.

Specification and configuration of the solar PV system

A hypothetical 10 MW solar PV plant is considered in this study. This size is based on three aspects: (i) 5MW being the minimum size for utility-scale solar PV plants (Al Garni & Awasthi, 2017) (Yimen & Dagbasi, 2019); (ii) the IEA's average utility-scale solar PV size of 17 MW for emerging markets (Bellini, 2019) (IEA, 2019); and (iii) Cameroon's recent drive to upscale solar PV adoption such as the planned development of the Maroua (15 MW) and Guider (10 MW) solar projects in the Far North of the country. The envisioned 10MW utility-scale solar PV plant is grid-connected with no storage. The solar PV configuration is depicted in Fig. 2.

The main solar PV plant components are the PV modules and inverter. In this study, the PV module of choice is the 345W monocrystalline silicone SPR-X21-345-COM module manufactured by SunPower (SunPower, 2017).

Efficiencies for current utility scale solar PV inverters average 98% according to the National Renewable Energy Laboratory (NREL) (Feldman et al., 2020). An inverter loading ratio of 1.34 is assumed in this study to minimize energy losses and maximize project economics. This

Table 1 PV modules specifications

Component	Specification	Specification
PV module	Module type	monocrystalline silicone
	Manufacturer	SunPower Corporation (Maxeon Solar)
	Series	SPR-X21-345-COM
	Nominal power	345W
	Maximum system voltage	600V
	Open Circuit Voltage (Voc)	68.2V
	Short circuit current (Isc)	6.39A
	Module efficiency	21.5%
	Operating temperature	-40 to 85
	Area	1.54m ²
Inverter	Weight	18.6kg
	Efficiency	98%
	Capacity	7.5MW

is also in accordance with recommendations by NREL for utility scale solar PV projects (Feldman et al., 2020). The solar PV module and inverter specifications are presented in Table 1.

The PV modules in our hypothetical system are assumed to be mounted on a single-axis tracking system, tilted at 6° facing the south to Ayompe and Duffy (2014) (Jacobson & Jadhav, 2018). According to RETScreen, the 10MW solar capacity requires approximately 29,000 solar PV modules spread over a surface area of 46535m².

Description of RETScreen

The RETScreen Clean Energy Management Software, generally shortened to RETScreen (or RETScreen Expert) is primarily used for identifying, assessing and optimizing technical and financial viability of potential clean energy projects. First launched in 1998, It has been actively developed and managed mainly by CANMET Energy Technology Centre located in Varennes (CETC-Varennes), a research center of the Natural Resources department of the Government of Canada (NRCan) (Leng et al., 1996) (OPENEI, 2022).

RETScreen integrates a series of databases (including NASA climate, product, cost, hydrology and databases) and advanced algorithms under an intelligent decision-making framework (NASA, 2022). RETScreen's capability includes evaluating energy production and savings, costs, emission reductions, financial viability and risk analysis of energy efficient technologies, energy performance, portfolio analysis and the full range of power generation technologies. Power generation includes conventional (gas turbines and reciprocating engines), renewables (solar, wind, tidal, hydro, geothermal) and cogeneration technologies (OPENEI, 2022) (RETScreen Innovation Lab, 2022).

RETScreen uses average monthly data, rather than average hourly data as required in other solar PV analysis models such as PVSIS and Homer. Despite its simplicity and ease of use, it incorporates enough detail to provide accurate results, with errors negligible in comparison to similar software (Psomopoulos et al., 2015). RETScreen has been applied in Cameroon (Yimen et al., 2018), Nigeria (Owolabi et al., 2019) and (Failed, 2018), in Bangladesh (Failed, 2016) and in Pakistan (Mehmood et al., 2014). In this study, RETScreen is used as a technoeconomic and environmental tool to assess some evaluation criteria (NPV, GHG emissions and capacity factor) for utility scale solar PV at alternative sites. RETScreen Version 8 is used in this study.

Climate data for each study location was gotten from the NASA climate database through RETScreen and from the NREL, the latter manually added. The fuel type in the business-as-usual scenario consisted of the current power generation mix Cameroon. Total electricity generation was 8043 GWh, constituting 65.7% from renewables (hydropower and solar PV), 16.6% from natural gas, 12.4% from off-grid and backup generators (diesel and gasoline-fired) and 5.3% from oil plants (IFC, 2019). This resulted in an Emission factor of 0.273 for Cameroon. Capital costs, and operation and maintenance costs of the utility- scale solar PV system were sourced from Allington et al. (2021) (Ayuketah et al., 2022). The summary of the technoeconomic inputs in RETScreen is presented in Table 2.

Table 2 RETScreen technoeconomic assumptions

Parameter	Value
Projective lifetime	25 years
Discount rate	10%
Inflation rate	2%
Debt interest rate	12%
Debt ratio	70%
Debt term	15 years
System PV Capital costs	1378 \$/kW
System O&M costs	17.91 \$/kW/yr
Electricity export rate	0.12\$/kWh
Transmission and distribution losses	10%
Transmission line costs	5000 \$/km
Inverter capital costs	3900/kW
Inverter lifetime	15yrs
Inverter replacement costs	80 \$/kW
Miscellaneous losses	5%

The AHP-TOPSIS process

The analytical hierarchy process (AHP) method

The AHP method was originally developed in the 1970's by Thomas Saaty (Saaty, 1980). It allows the combination of both qualitative and quantitative inputs in identifying the preferred or ranking of alternatives. AHP is one of the most comprehensive and widely used MCDM methods, given its effectiveness in handling multiple and sometimes conflicting objectives (Al Garni & Awasthi, 2017). The AHP method accounts for inconsistency in the decision maker's evaluations to determine how robust the AHP technique results could be a basis for decision making. AHP has been applied in utility-scale solar PV site selection process, including (Al Garni & Awasthi, 2017) (Yimen & Dagbasi, 2019). In this study, AHP is used for calculating weights of criteria for siting the solar PV plant.

The AHP method essentially breaks down the large decision problem into smaller sub-problems, structuring them into hierarchical levels. The uppermost level represents the primary objective, the middle level represents the criteria influencing the decision and the bottom levels represent the alternatives. The decision maker assesses each level element in a pairwise comparison judgement to express relative impact or strength against their parent node. These pairwise comparisons judgements are interpreted by assigning integer values, which are used to form a decision matrix, and the resulting criteria weights. To enable the sifting of decision makers' inconsistent judgment during the pairwise comparison process, a consistency ratio (CR) is utilized.

The AHP hierarchical structure shows the relationship between each level, while assigning weights to each

Table 3 Preference scores and interpretation (Saaty, 1980)

a_{ij} value	Interpretation
1	Criterion i and criterion j are of equal importance
3	Criterion i is slightly more important than criterion j
5	Criterion i is moderately more important than criterion j
7	Criterion i is strongly more important than criterion j
9	Criterion i is absolutely of more importance than criterion j
2, 4, 6, 8	Intermediate values. For example, a value of 4 means that criteria i is slightly of more importance than j , but not moderately more important (midway)

Table 4 Random consistency index for $n = 12$ criteria (Al Garni & Awasthi, 2017)

Number of criteria (n)	2	3	4	5	6	7	8	9	10	11	12
R1	0	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49	1.51	1.48

decision-making criterion. The AHP process steps for n number of criteria is:

i. Construct a pairwise comparison matrix, P

For n number of criteria, $P = (n \times n)$, p_{ij} is the entry in the i th row and the j th column of P . p_{ij} denotes the preference score i.e., how much more important criterion i is to j with respect to the alternative. The preference score represents a translation into numerical quantities of the decision maker's verbal judgments. The 1–9 integer scale proposed by Saaty is the most widely used (Table 3) (Saaty, 1980). The entries of A must satisfy the constraints in Eq. (1)

$$p_{ij} \cdot p_{ji} = 1 \quad (1)$$

ii. Establish the normalized pairwise comparison matrix, $\bar{P}$

This entails making sum of each column to be equal to 1 using Eq. 2 to compute $\bar{p}_{ij}$ for each preference score in $\bar{P}$

$$\bar{p}_{ij} = \frac{p_{ij}}{\sum_{i=1}^n p_{ij}} \quad (2)$$

iii. Obtain relative weights, w_i

This involves computing the average across rows to obtain the relative weights of the criteria using Eq. 3.

$$w_i = \frac{\sum_{i=1}^n \bar{p}_{ij}}{n} \quad (3)$$

The relative weight for each criterion is in the range 0–1. The higher the relative weight, the more influencing the criteria in decision making, the utility-scale solar PV plant in this study.

iv. Evaluate the consistency ratio (CR).

The CR is calculated using Eq. 4.1, where CI is the consistency index. RI is the random consistency index, determined by the number of criteria. RI is presented in Table 4.

CI is calculated from Eq. 4.2 where λ_{max} is the comparison matrix's maximum eigenvalue.

$$CI = \frac{\lambda_{max} - n}{(n - 1)} \quad (4.1)$$

$$CR = \frac{CI}{RI} \quad (4.2)$$

The CR indicates the decision makers consistency during the pairwise process. The smaller the consistency ratio, the more useful the AHP weights can be applied for decision making. However, a $CR \leq 0.10$ is considered satisfactory for decision making. Otherwise, serious inconsistencies may exist, and the results of the AHP process may be inappropriate for decision making.

The TOPSIS method

The TOPSIS method is based on the concept that the chosen alternative must have the shortest geometric distance from the positive ideal solution and the farthest geometric distance from the negative-ideal solution. It was first developed by Yoon and Hwang in 1981 (Huang & Yoon, 1981). The TOPSIS method assumes each criterion has a monotonically increasing or decreasing preference which makes it easier to identify the best-ideal and worst-ideal solutions. Consequently, the preference order is generated through a comparison of their geometric distances, usually their Euclidean distances. The TOPSIS method has been applied for utility PV selection by Akçay and Atak (2018) (Nazari et al., 2018).

The TOPSIS method comprises the following steps: (Huang & Yoon, 1981) (Mateo, 2012).

(1) Compute the normalized decision matrix.

For n criteria and m alternatives, the decision matrix is first formulated, then normalized for each entry using Eq. 5

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{j=1}^n x_{ij}^2}} \quad (5)$$

where x_{ij} is the value of the i th criteria for the alternative A_j ; ($i = 1, 2, \dots, n$; $j = 1, 2, \dots, m$)

(2) Compute the weighted normalized decision matrix.

This is calculated using Eq. 6

$$v_{ij} = w_i r_{ij} \quad (6)$$

where w_i is the weight of the i th criterion calculated from Eq. 3, and $\sum_{i=1}^n w_i = 1$.

(3) Calculate the positive-ideal and negative-ideal solutions.

The positive ideal (A^+) and negative-ideal (A^-) solutions are calculated, respectively, using Eq. 7.1 and Eq. 7.2

$$A^+ = \{v_1^+, \dots, v_n^+\} \\ = \{(max_j v_{ij} | i \in I'), (min_j v_{ij} | i \in I'')\} \quad (7.1)$$

$$A^- = \{v_1^-, \dots, v_n^-\} = \{(min_j v_{ij} | i \in I'), (max_j v_{ij} | i \in I'')\} \quad (7.2)$$

where I' and I'' are, respectively, the associate benefit or detriment criteria.

(4) Compute separation distances

The alternative distances from the positive-ideal (D^+) and negative-ideal (D^-) solutions are calculated respectively, using Eq. 8.1 and Eq. 8.2

$$D_j^+ = \sqrt{\sum_{i=1}^n (v_{ij} - v_i^+)^2} \quad (8.1)$$

$$D_j^- = \sqrt{\sum_{i=1}^n (v_{ij} - v_i^-)^2} \quad (8.2)$$

(5) Compute the closeness to positive-ideal solution.

The relative closeness of alternative a_j to the positive-ideal solution is calculated using Eq. 9

$$C_j^+ = \frac{D_j^-}{(D_j^+ + D_j^-)} \quad (9)$$

(6) Rank the alternatives.

The value C_j^+ of each alternative are sorted according to magnitude, then arranged in decreasing order. The highest ranked alternative is the best solution.

Case study

Overview of Cameroon and the southern interconnected grid

Cameroon is a central African nation with a population of 25.9 million and a national electrification rate of 70% in 2019. Rural and urban electrification rates were respectively 32% and 98%. As of July 2020, the official installed capacity was 1541.4MW (Ayuketah et al., 2023b). Large hydropower made up 61% while thermal (natural gas and oil) capacity constituted 38%. Despite a total RES installed capacity of 14.5MW (solar PV, 14.2MW and small hydropower, 0.3MW), its share in the total generation mix is meagre, approximately 1%. Apart from an eternal generation inadequacy in its power system, a dilapidated grid infrastructure, overloaded transmission and distribution network, in addition to high power losses are some of the issues of the grid (Ayuketah et al., 2023a) (Ayuketah et al., 2022).

The issues of the power system result in erratic power supply and recurrent power outages which damage appliances. In the industry sector, power outages resulted in estimated output losses between 16 and 50% for 80% of firms. Enforced back-up gensets operation for these firms costing over four times grid rates (Ayuketah et al., 2023b) (Diboma & Tatietse, 2013). The location-constrained nature of large hydropower resources in Cameroon, a priority for the state, complicates the grid decentralization process necessary to resolve some of these challenges. The state has earmarked uptake of RES as a start to resolving the generation inadequacy problem and transforming its power system (Ayuketah et al., 2023a).

The national grid system is divided into two independent areas: the northern part (covering three regions and southern parts (covering seven regions), respectively being served by the Northern Interconnected Grid (NIG) and the Southern Interconnected Grid (SIG). The area of study is the southern part which is served by the SIG. The choice is based on the SIG being the largest, covering seven of the ten regions of Cameroon. The southern area also includes the industrial and economic capital of Douala, and Yaoundé the political capital. Furthermore, SIG holds over 93% of installed capacity in the country. Finally, the area has sufficient solar resource, with irradiation averaging 4.9 kWh/m² over the national territory (Sebastian et al., 2014). The southern area also has adequate potential locations for utility scale solar PV plants. The description of the study area is depicted in Fig. 3.

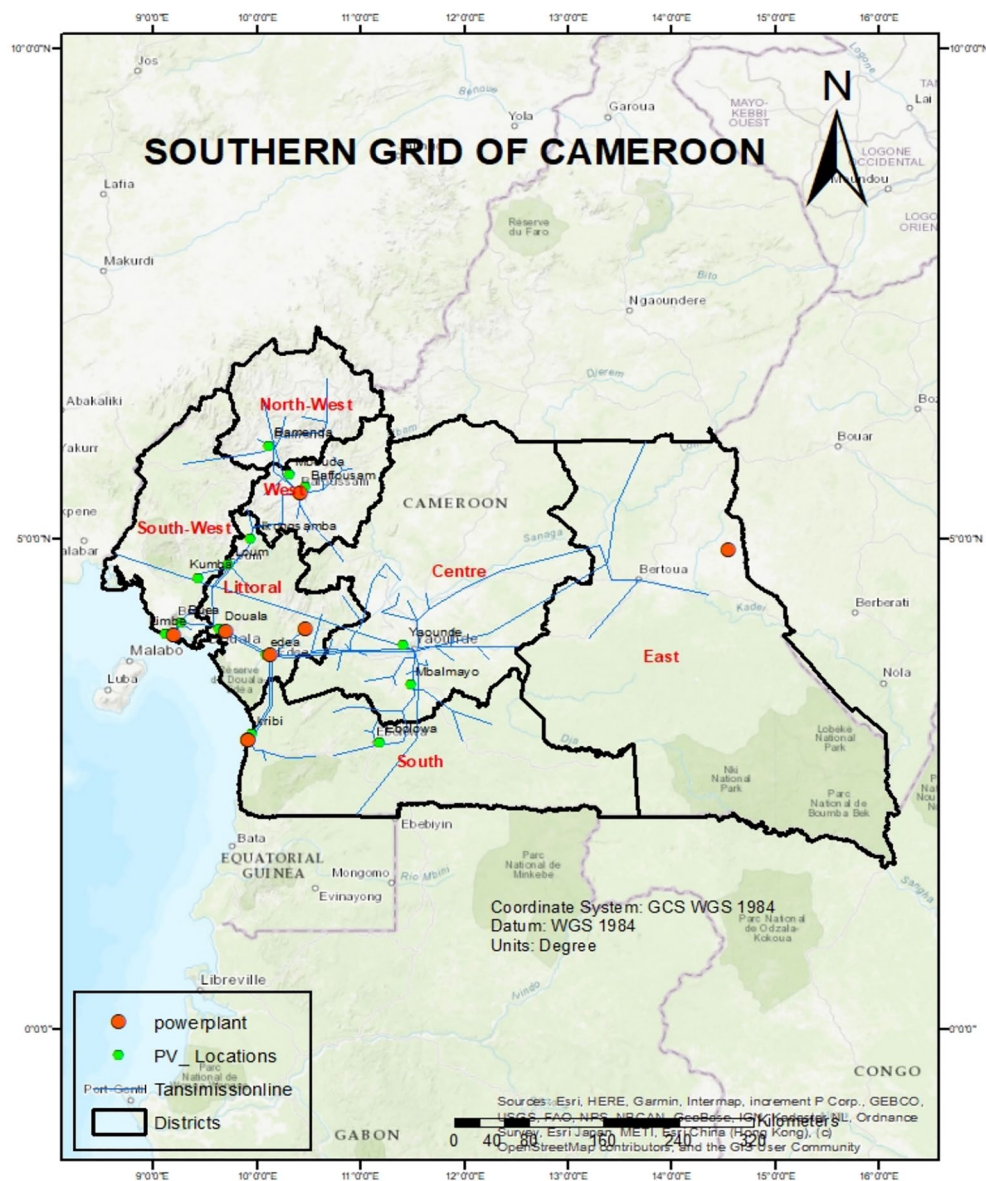


Fig. 3 The southern part of Cameroon

Calculation of criteria weights

Based on the extensive evaluation of criteria weights by Deveci et al. (Deveci et al., 2021), similar studies by Al Garni and Awasthi (2016) (Akçay & Atak, 2018) (Kaya & Kahraman, 2011) (Colak et al., 2020) and author's knowledge of the Cameroon electricity system landscape, the pairwise comparison matrix was developed, as presented in Table 5.

The resulting criteria weights obtained from pairwise comparisons are depicted in Fig. 4. To verify the consistency of the pairwise comparison process, the

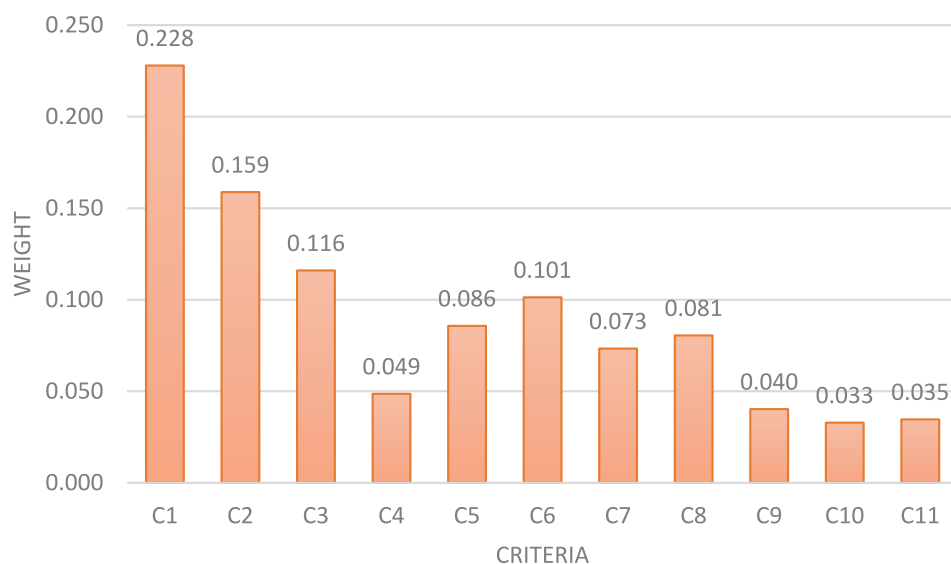
Consistency ratio was calculated to be $CR = 0.07$. This is less than 0.10, thus the value judgements in Table 3 could be used as basis for decision making.

Description of alternative sites

Fourteen locations were selected considering the capitals of each region, followed by population size of towns in the SIG (Population of Cities in Cameroon, 2022). The sites were selected considering the exclusion criteria identified in Shao et al. (2020). Thus, the appropriate buffer was applied in each case.

Table 5 The pairwise comparison matrix of the eleven decision criteria

Criteria	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11
C1	1	2	2	4	3	4	4	5	4	6	4
C2	1/2	1	2	4	4	2	2	3	2	3	5
C3	1/2	1/2	1	2	3	2	2	1/2	3	4	3
C4	1/4	1/4	1/2	1	1/2	1/4	1/3	1/4	2	3	2
C5	1/3	1/4	1/3	2	1	1	1	3	3	4	2
C6	1/4	1/2	1/2	4	1	1	2	2	2	7	2
C7	1/4	1/2	1/2	3	1	1/2	1	2	1	3	2
C8	1/5	1/3	2	4	1/3	1/2	1/2	1	3	2	2
C9	1/4	1/2	1/3	1/6	1/3	1/2	1	1/3	1	1	1
C10	1/6	1/3	1/4	1/3	1/4	1/7	1/3	1/2	1	1	2
C11	1/4	1/5	1/3	1/2	1/2	1/2	1/2	1/2	1	1/2	1

**Fig. 4** Description of criteria weights

ArcGIS was used to measure the proximity to transmission lines, protected area and water way. Vector data of power lines were gotten from Cameroon Electricity Transmission Network, (2020), protected areas from Protected Areas-Vector, 2022, roads from Cameroon - Roads, 2022, and water ways from Cameroon major rivers, 2022. The alternative locations are presented in Table 6.

Sensitivity analysis

A sensitivity analysis is necessary to evaluate the robustness of the results against the subjectivity in the pairwise comparison process, or how a change in put

parameters influence the ranking of alternative sites. In this study, the sensitivity analysis conducted considers the effect of changes in criteria weights on the entire suitability index. To this end, four cases are examined:

- *Case 1:* Equal weights for all criteria.
- *Case 2:* Technical criteria the only basis for decision making: i.e., weights equally distributed among the technical criteria, with other criteria weights assumed zero.
- *Case 3:* Project economics is the only basis for decision making, i.e., weights equally distributed among the economic criteria with other criteria weights assumed zero. Two sub-cases are considered: (i)

Table 6 Location and attributes of alternative sites

Name	Longitude (N)	Latitude °	Elevation (m)	AYT (°C)	ADSR (kWh/m ² /d)	AYWS (m/s)	PTL (m)	PRA (m)	PTR (m)	PPA (m)	PWW (m)	Land use
Baffousam	10.47	5.53	1374	22.6	4.98	2.4	6689	1500	1700	54,190	6648	1
Bamenda	10.11	5.95	1223	20.7	4.96	1.7	1324	820	1263	30,209	11,118	3
Buea	9.27	4.14	618	23.9	4.4	2.3	388	920	537	7902	3030	3
Douala	9.63	4.08	882	24.9	4.28	1.7	567	912	620	5814	213	2
Ebolowa	11.18	2.93	621	22.8	4.63	0.7	2575	1537	1840	50,684	7171	2
Edea	10.09	3.81	41	25	4.42	1.4	883	1025	1231	2790	2130	2
Kribi	9.96	3.01	42	25.8	4.57	3	1358	640	580	26,904	883	2
Kumba	9.44	4.60	198	23.9	4.5	2.3	3512	1724	1858	6703	1171	2
Limbe	9.12	4.03	160	23.9	4.07	2.3	8975	1358	954	3323	2164	3
Loum	9.73	4.73	350	23.9	4.49	2.3	846	1012	1145	3686	4231	2
Mbalmayo	11.49	3.51	669	23	4.5	1.9	1688	740	956	11,495	1393	2
Mbouda	10.32	5.65	1256	22.6	4.85	2.4	1058	1289	2477	36,238	3841	3
Nkongsamba	9.93	5.00	1017	23.9	4.68	2.3	272	950	292	5384	4887	2
Yaounde	11.41	3.91	787	23	4.67	1	1327	912	915	36,107	1157	2

AYT average yearly temperature, ADSR average daily solar radiation, AYWS average yearly wind speed, PTL Proximity to transmission lines, PRA Proximity to residential areas, PTR Proximity to road, PPA Proximity to protected areas (forest, wildlife, parks, etc.) PWW Proximity to water way, LU land use (1 for Bare land, 2 for Forest, and 3 Farm land)

Table 7 RETScreen results at alternative sites

Name	Energy (MWh/yr)	CF (%)	GHG (tCO ₂ /year)	NPV (million USD)	SPP (Years)
Baffousam	15.68	17.9	4286	16.84	4.7
Bamenda	15.78	18.0	4314	17.06	4.6
Buea	13.88	15.8	3793	12.94	5.3
Douala	13.43	15.3	3670	11.97	5.5
Ebolowa	14.6	16.7	3991	14.5	5
Edea	13.85	15.8	3785	12.87	5.3
Kribi	14.222	16.2	3888	13.69	5.2
Kumba	14.17	16.2	3873	13.52	5.2
Limbe	12.87	14.7	3518	10.76	5.8
Loum	14.13	16.1	3863	13.49	5.2
Mbalmayo	14.21	16.2	3884	13.66	5.2
Mbouda	15.32	17.5	4187	16.05	4.8
Nkongsamba	14.724	16.8	4024	14.77	5
Yaounde	14.71	16.8	4020	14.73	5

CF capacity factor, GHG avoided GHG emissions, NPV net present value, SPP simple payback period

weights equally distributed among the economic criteria (Case 2a); and (ii) weights equally distributed between the economic criteria bar land use (Case 2b).

- *Case 4:* Environmental criteria is the only basis for decision making: i.e., weights equally distributed among the environmental criteria, with other criteria weights assumed zero.

Results and discussion

RETScreen results

The RETScreen simulation results of the fourteen alternative locations are presented in Table 7.

The utility-scale solar PV project NPV's range from \$10.76 million in Limbe, to \$17.06 million in Bamenda, with a mean of \$14.06 million (Table 7). If the NPV was the sole basis for selection, the three best locations would be Bamenda, Baffousam, and Mbouda (in descending order). of the ranking of these locations is substantiated as they still have the best SPP, energy generated, CF and avoided GHG emissions. Curiously, these locations are situated in the Cameroon highlands (North West and West regions). In contrast, the lowest NPVs are in Limbe, Douala and Edea (in ascending order), which are all along the coast. Thus, the Cameroon high lands are the most economically profitable locations for construction of a utility-scale solar PV projects in the geographic area of the SIG. Nevertheless, all fourteen locations have $SPP \leq 6\text{years}$, meeting the criteria by Yimen and Dagbasi

Table 8 Preference scores and ranking of alternative locations

Name	Preference score	Rank
Baffousam	0.989	1
Nkongsamba	0.505	2
Yaoundé	0.505	3
Ebolowa	0.504	4
Kribi	0.499	5
Mbalmayo	0.499	6
Kumba	0.499	7
Loum	0.498	8
Edea	0.495	9
Douala	0.49	10
Bamenda	0.161	11
Mbouda	0.139	12
Buea	0.066	13
Limbe	0.002	14

(2019) which indicates the alternative locations are all economically viable.

Ranking based on AHP-TOPSIS

Based on the pairwise comparison, the ranking of alternative sites is in the order: (1) Baffousam; (2) Nkongsamba; (3) Yaoundé; (4) Ebolowa; (5) Kribi; (6) Mbalmayo; (7) Kumba; (8) Loum; (9) Edea; (10) Douala; (11) Bamenda; (12) Mbouda; (13) Buea; and (14) Limbe. This is represented in Table 8.

Table 9 Sensitivity results

Location	Ranking				
	Case 1	Case 2	Case 2		Case 3
			Case 2a	Case 2b	
Baffousam	1	2	1	2	3
Bamenda	11	1	11	1	1
Buea	13	11	13	11	7
Douala	10	13	10	13	14
Ebolowa	4	6	4	6	2
Edea	9	12	9	12	8
Kribi	5	7	5	7	13
Kumba	7	9	7	9	12
Limbe	14	14	14	14	9
Loum	8	10	8	10	5
Mbalmayo	6	8	6	8	11
Mbouda	12	3	12	3	6
Nkongsamba	2	4	2	4	4
Yaounde	3	5	3	5	10

Despite Bamenda having the highest energy generated and SPP, we observed that the influence of other criteria lowers its preference to 11th. However, Limbe, which was last, remains the least preferable option. Thus, the simultaneous consideration of all influencing criteria is important for the optimal selection or location of renewable energy projects.

Sensitivity results

The results of the sensitivity analysis are presented in Table 9.

Generally, Baffousam is the best location overall from equal consideration of criteria (Case 1). This is followed by Nkongsamba and Yaoundé. In contrast, Bamenda is the best location if decision making was based on either the technical criteria (Case 2) or environmental criteria (Case 3), despite its 11th position in the economic criteria. On average, however, Edea is the best location under all cases, followed by Kumba and Loum.

The influence of land use criterion is observed from Case 2. Despite Bamenda having the best NPV and SPP values in the economic criteria, the influence of land use makes it rank 11th (Case 2a). However, an assumption of equal land attainability resets this ranking, pushing it to 1st (Case 2b). This highlights the effect of land use on project economics, and the overall impact of utility-scale PV site selection.

Discussion

This study integrates technical, economic and environmental criteria with the time value of money, a critical gap in existing frameworks (Al Garni & Awasthi, 2016) (Al Garni & Awasthi, 2017) (García-Cascales & Sánchez-Lozano, 2013). By embedding RETScreen-derived financial indicators (e.g., NPV, SPP) into the MCDM process, the model explicitly accounts for project viability, offering a more realistic decision-making tool for utility-scale solar PV site selection. Unlike prior studies, the framework is modular, i.e., stakeholders can adapt, exclude, or expand criteria (adding social factors for example) to align with their priorities, whether for public utilities, private investors, or policy makers. This adaptability is vital for Cameroon which aim to expand solar capacity to 700MW by 2035 (MINEPDD, 2016), and provide actionable insights for prioritizing high potential sites like Baffousam within the SIG.

The findings of this study provide actionable insights for energy planners and policymakers in Cameroon and similar regions. Baffousam, identified (based on pairwise comparison) as the optimal site for utility-scale solar PV deployment offers high solar potential, proximity to the grid, and favorable land use conditions. Prioritizing Baffousam aligns with Cameroon's Vision 2035 industrialization goals, while reducing transmission losses and enhancing grid reliability (Ayuketah et al., 2024) (Ayuketah et al., 2022). Furthermore, Policymakers must balance solar PV deployment with agricultural and ecological priorities as land use criteria significantly influence economic viability as in the case of Bamenda in the sensitivity analysis. Lastly, Stakeholder engagement is critical to address the inherent subjectivity of AHP-based weighting, ensuring consensus among local communities, investors, and technical experts (Shao et al., 2020) (Kaya & Kahraman, 2011).

However, the study has limitations. Firstly, RETScreen's reliance on historical weather data may not account for future climate variability, necessitating dynamic modeling with climate projections. Secondly, while AHP pairwise comparisons were validated (consistency ratio < 0.1), inherent subjectivity persists. Future studies could integrate Delphi methods or fuzzy (like in Rekik & Alimi, 2024a) to refine weights. Third, static assumptions (e.g., fixed financial parameters) limit dynamic analysis. Stochastic frameworks incorporating real-time economic and uncertainty considerations are recommended. Finally, the geographical focus of this study excludes off-grid regions which hold all the nation's current installed solar PV capacity (Ayuketah et al., 2023a) (Ayuketah

et al., 2023b), and where hybrid systems (solar-wind-storage) could holistically address energy poverty is a disadvantage. Future work should expand the framework to these regions and explore hybrid renewable energy systems to support universal energy access and resilience as in Rekik and Alimi (2023).

Conclusion

Renewable energy resources, particularly solar PV can play a significant role in decentralizing electricity access and mitigating GHG emissions. This study presents an original decision-making framework that integrates the time-value of money into the site selection process for utility-scale solar PV projects. By combining GIS, RETScreen's financial metrics and MCDM methods, the framework hopes to bridge the gap between traditional energy planning tools and modern sustainability evaluations. GIS streamlined preliminary site screening, reducing time and costs for stakeholders, while RETScreen embedded critical financial viability indicators into the analysis.

The framework was applied to fourteen sites Cameroon's SIG, prioritizing technical, economic, and environmental criteria. AHP assigned weights to eleven chosen criteria, validated for consistency, and TOPSIS ranked sites based on their proximity to ideal solutions. Bafousam emerged as the optimal location due to its high solar potential, favorable land conditions and proximity to existing grid infrastructure. However, ranking remain sensitive to stakeholder priorities, underscoring the need for participatory decision-making. Deploying solar PV in prioritized locations would reduce transmission losses, enhance grid reliability, and support Cameroon's Vision 2035 goals. While the study advances renewable energy planning, it excludes social criteria and assumes static technoeconomic parameters. Future work should incorporate dynamic modelling, expand to off-grid regions and consider hybrid renewable systems. This framework offers a replicable blueprint for sub-Saharan Africa, harmonizing financial viability with socio-environmental priorities to accelerate the clean energy transition.

Author contributions

Author Contributions: Conceptualization, Y.A., M.E. and C.M.; Methodology, Y.A.; Writing -original draft, Y.A.; Writing—review & editing, Y.A., M.E., C.M.; Supervision, M.E. and C.M.; All authors have read and agreed to the published version of the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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